

Solutions to Problem Set 2

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Problem 1

1. Solving the system $\bar{x} = A\bar{x}$, we get a unique steady state $\bar{x} = 0$. Next, find the eigenvalues of matrix A :

$$|A - \lambda I| = (1 - \lambda)(1.5 - \lambda) - 0.5 = \lambda^2 - 2.5\lambda + 1 = 0 \quad \Rightarrow \quad \Lambda = \begin{pmatrix} 2 & 0 \\ 0 & 0.5 \end{pmatrix}.$$

It follows that the steady state is a saddle point.

2. The isolines are given by

$$\Delta x_{1t+1} = -0.5x_{2t}$$

$$\Delta x_{2t+1} = -x_{1t} + 0.5x_{2t}$$

3. Solve for the eigenvectors:

$$A - \lambda_1 I = \begin{pmatrix} -1 & -0.5 \\ -1 & -0.5 \end{pmatrix} \quad \Rightarrow \quad v^1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$A - \lambda_2 I = \begin{pmatrix} 0.5 & -0.5 \\ -1 & 1 \end{pmatrix} \quad \Rightarrow \quad v^2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

It follows that $A = Q\Lambda Q^{-1}$, where

$$Q = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}, \quad Q^{-1} = \frac{1}{3} \begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}.$$

4. From $y_t = Q^{-1}x_t$ and $y_t = \Lambda^t y_0$ we get

$$y_0 = \frac{1}{3} \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad y_1 = \frac{1}{3} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad y_2 = \frac{1}{3} \begin{pmatrix} 4 \\ 0.5 \end{pmatrix},$$

$$x_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 1.5 \\ -2.5 \end{pmatrix}.$$

Problem 2

1. Let λ_t denote the co-state. Then the Hamiltonian can be written as

$$\mathcal{H}_t = e^{-\rho t} \left(\alpha C_t - \frac{1}{2} C_t^2 \right) + \lambda_t (r B_t + Y - C_t)$$

and the principle of maximum implies

$$0 = \frac{\partial \mathcal{H}_t}{\partial C_t} = e^{-\rho t} (\alpha - C_t) - \lambda_t$$

$$-\dot{\lambda}_t = \frac{\partial \mathcal{H}_t}{\partial B_t} = r \lambda_t$$

2. Combining the optimality conditions, obtain

$$\rho e^{-\rho t} (\alpha - C_t) + e^{-\rho t} \dot{C}_t = r e^{-\rho t} (\alpha - C_t),$$

which can be simplified to get the Euler equation

$$\dot{C}_t = (r - \rho)(\alpha - C_t).$$

The transversality condition is given by

$$\lim_{t \rightarrow \infty} \lambda_t B_t = \lim_{t \rightarrow \infty} e^{-\rho t} (\alpha - C_t) B_t = 0.$$

3. The evolution of C_t and B_t is described by a system $\dot{x}_t = A x_t$, where

$$x_t = \begin{pmatrix} B_t \\ C_t \end{pmatrix}, \quad A = \begin{pmatrix} r & -1 \\ 0 & -(r - \rho) \end{pmatrix}.$$

As long as $r \neq \rho$, matrix A is non-singular, the steady state is unique and is given by $\bar{C} = \alpha$, $\bar{B} = \frac{1}{r}(\alpha - Y)$. Otherwise, there is continuum of steady states.

4. Assuming $\rho \neq r$, notice first that the iso-lines are given by

$$\begin{aligned}\dot{B}_t = 0 &\Leftrightarrow C_t = rB_t + Y, \\ \dot{C}_t = 0 &\Leftrightarrow C_t = \alpha.\end{aligned}$$

Find the eigenvalues and the eigenvectors of matrix A :

$$\Lambda = \begin{pmatrix} r & 0 \\ 0 & -(r - \rho) \end{pmatrix}, \quad Q = \begin{pmatrix} 1 & 1 \\ 0 & 2r - \rho \end{pmatrix}.$$

When $r > \rho$, there is one positive and one negative eigenvalue and the steady state is a saddle point. When $r < \rho$, both eigenvalues are positive and the steady state is a source.

Problem 3

1. From the previous problem set, it follows

$$\begin{aligned}Q_t &= 1 + \phi \frac{I_t}{K_t}, \\ Q_t &= \beta \left(\alpha A K_{t+1}^{\alpha-1} + \frac{\phi}{2} \frac{I_{t+1}^2}{K_{t+1}^2} \right) + \beta Q_{t+1}.\end{aligned}$$

Substitute in the capital law of motion to substitute out investment:

$$\begin{aligned}Q_t &= 1 + \phi \frac{K_{t+1}}{K_t} - \phi, \\ Q_t &= \beta \left\{ \alpha A K_{t+1}^{\alpha-1} + \frac{\phi}{2} \frac{(K_{t+2} - K_{t+1})^2}{K_{t+1}^2} + Q_{t+1} \right\}.\end{aligned}$$

2. K_t is a state variable and Q_t is a control variable.

3. Impose $K_{t+1} = K_t = \bar{K}$ and $Q_{t+1} = Q_t = \bar{Q}$ to solve for the steady state:

$$\bar{Q} = 1,$$

$$\bar{K}^{1-\alpha} = \frac{\beta\alpha A}{1-\beta}.$$

4. Rewrite the equilibrium system in terms of log-deviations:

$$\bar{Q}(1 + q_t) = 1 + \phi(1 + k_{t+1} - k_t) - \phi,$$

$$\bar{Q}(1 + q_t) = \beta \left[\left(\alpha A \bar{K}^{\alpha-1} (1 - (1-\alpha)k_{t+1}) + \frac{\phi}{2}(k_{t+2} - k_{t+1})^2 \right) + \bar{Q}(1 + q_{t+1}) \right].$$

Using the steady-state values, get rid of the zero-order and second-order terms:

$$q_t = \phi(k_{t+1} - k_t),$$

$$q_t = \beta \left[-\alpha(1-\alpha)A\bar{K}^{\alpha-1}k_{t+1} + q_{t+1} \right].$$

Thus, the dynamic system can be written as $x_{t+1} = Bx_t$,

$$x_t = \begin{pmatrix} k_t \\ q_t \end{pmatrix}, \quad B = \begin{pmatrix} 1 & \frac{1}{\phi} \\ \kappa & \frac{1}{\beta} + \frac{\kappa}{\phi} \end{pmatrix},$$

where $\kappa \equiv \frac{(1-\alpha)(1-\beta)}{\beta}$ and the steady-state value of $\bar{K}$ has been used.

5. Define

$$f(\lambda) = |B - \lambda I| = \lambda^2 - \left(1 + \frac{1}{\beta} + \frac{\kappa}{\phi} \right) \lambda + \frac{1}{\beta}.$$

Notice that $f(0) > 0$ and $f(1) < 0$, which implies $\lambda_1 > 1$ and $0 < \lambda_2 < 1$. Thus, the number of eigenvalues greater than one is equal to the number of control variables and the Blanchard-Kahn condition is satisfied. For future reference

$$\lambda_{1,2} = \frac{1}{2} \left(1 + \frac{1}{\beta} + \frac{\kappa}{\phi} \pm \sqrt{\left(1 + \frac{1}{\beta} + \frac{\kappa}{\phi} \right)^2 - \frac{4}{\beta}} \right).$$

6. For eigenvalue λ , the corresponding eigenvector is

$$B - \lambda I = \begin{pmatrix} 1 - \lambda & \frac{1}{\phi} \\ \kappa & \frac{1}{\beta} + \frac{\kappa}{\phi} - \lambda \end{pmatrix} \Rightarrow v = \begin{pmatrix} 1 \\ \phi(\lambda_1 - 1) \end{pmatrix}.$$

It follows that

$$Q = \begin{pmatrix} 1 & 1 \\ \phi(\lambda_1 - 1) & \phi(\lambda_2 - 1) \end{pmatrix}, \quad Q^{-1} = \frac{1}{\phi(\lambda_2 - \lambda_1)} \begin{pmatrix} \phi(\lambda_2 - 1) & -1 \\ \phi(1 - \lambda_1) & 1 \end{pmatrix}.$$

Iterating forward the first equation of the decoupled system, we get $y_{1t} = 0$ and hence,

$$q_t = -\phi(1 - \lambda_2)k_t.$$

7. Substitute the optimal q_t into the first equation of the original system:

$$k_{t+1} = \lambda_2 k_t.$$

Given that $\lambda_2 \in (0, 1)$, capital stock monotonically converges to the steady state level.

8. The isolines are given by

$$\begin{aligned} \Delta k_{t+1} &= \frac{1}{\phi} q_t, \\ \Delta q_{t+1} &= \kappa k_t + \left(\frac{1}{\beta} + \frac{\kappa}{\phi} - 1 \right) q_t. \end{aligned}$$